DATA ANALYTICS PORTFOLIO PROJECT

E-Commerce Sales & Profitability Analytics

Where an online retailer makes and loses money across products, regions, discounts and returns.

MySQL | Python | Power BI

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PROJECT SCOPE

A sample store, from raw tables to decisions

1,000	82	5,000	12,558
customers	products	orders	order line items

Jan 2024 – Jul 2026 · 5 categories · 4 regions · 5 payment methods. Revenue and profit are counted on **Delivered** orders only.

WORKFLOW

Five steps, one pipeline

1

MySQL

5 related tables with keys and constraints

2

Clean

Types, nulls, duplicate keys, impossible values, joins

3

Explore

8 business questions on sales, profit and returns

4

Test

Welch t-test and a Random Forest profit model

5

Present

Power BI KPIs, trends and slicers

Tools: MySQL · pandas · NumPy · Matplotlib · Seaborn · SciPy · scikit-learn · Power BI

HEADLINE NUMBERS

A healthy 34% margin on 153M revenue

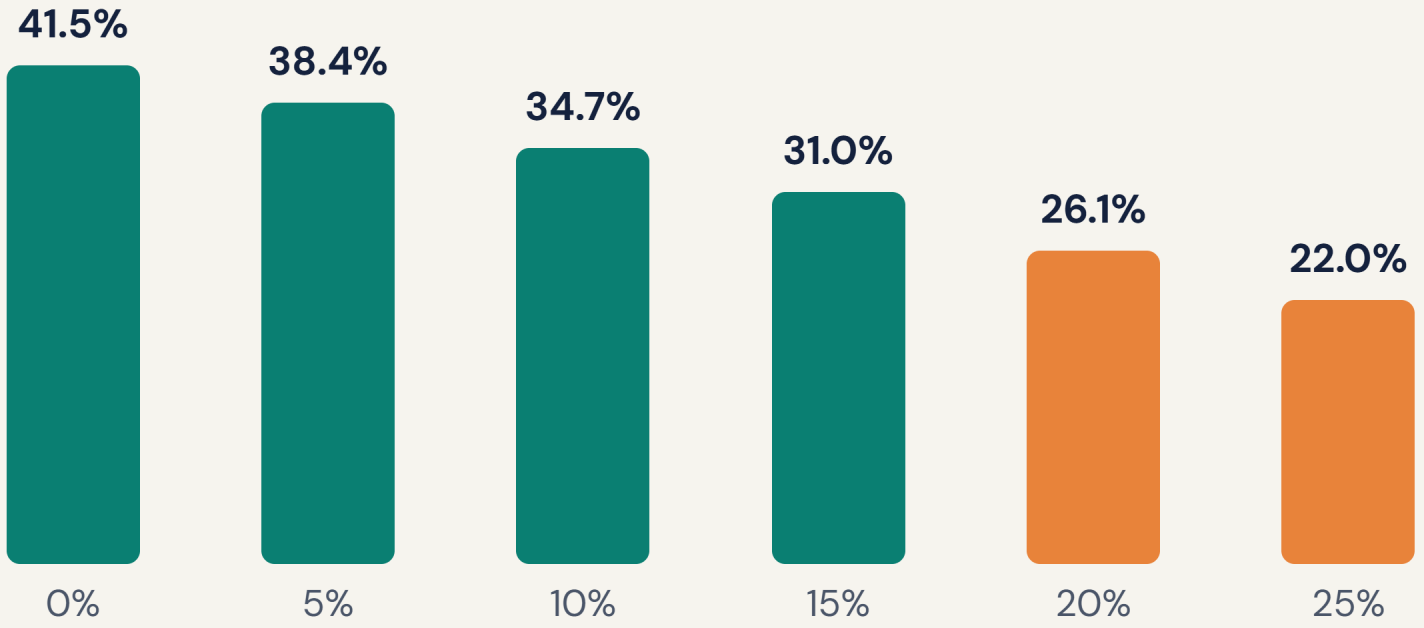
153.3M	52.5M	34.3%	4,386	34,955
revenue	profit	profit margin	orders	avg order value



Order outcomes: **87.7% delivered** · 7.1% cancelled · 5.1% returned

FINDING 1 · DISCOUNTS

Deep discounts erode margin



120

loss-making sales

Every one of them happened at a 20% or 25% discount.

Profit margin by discount level, Delivered orders (bars show margin; x-axis is the discount).

FINDING 2 · CATEGORIES

Revenue rank is not profit rank

Category	Revenue	Profit	Margin
Home	41.7M	13.9M	33.4%
Electronics	39.5M	13.7M	34.8%
Fashion	30.5M	10.6M	34.8%
Beauty	24.9M	7.9M	31.6%
Books	16.7M	6.4M	38.2%

38.2%

Books earn the best margin

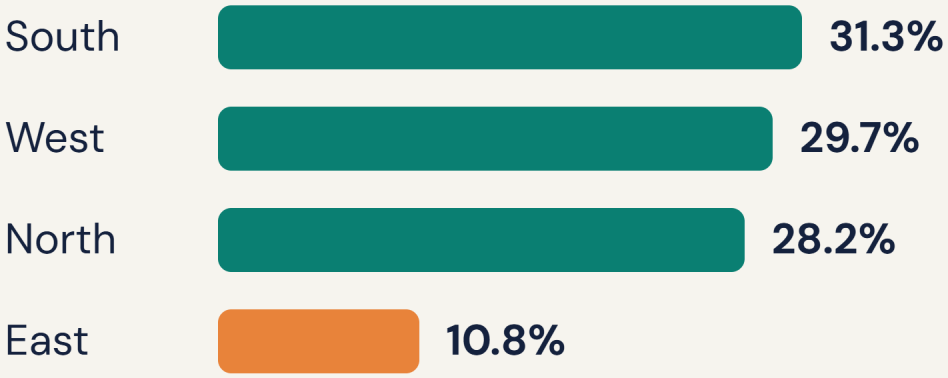
Beauty earns the lowest at 31.6%, despite ranking fourth on revenue.

Delivered orders only. Home and Electronics bring the most profit in absolute terms.

FINDINGS 3 & 4 · REGIONS AND TREND

East lags, and growth has stalled

Share of revenue by region



Margins are similar (about 34%), so East is a volume problem, not a pricing one.

FULL-YEAR REVENUE

2024	2025
60.6M	58.0M

−4.4% year over year

Monthly revenue sits at about 4.5–5.5M with a dip every February.

FINDING 5 · RETURNS

Most returns are operational



5.1%

of orders were returned

A further 7.1% were cancelled. 257 returns in total.

Late delivery, damage and wrong items (about 60% of returns) can be reduced through carrier tracking and warehouse quality checks.

FINDING 6 · HYPOTHESIS TEST

Bigger orders, but not because of discounts

Per order	Discounted	No discount
Orders	4,230	156
Avg order value	35.6K	18.8K
Items per order	2.55	1.20
Value per item	13.9K	15.7K

WELCH T-TEST

$p < 0.001$, reject H_0

Discounted orders are worth more, but they also hold twice the items.

Takeaway: the gap reflects basket size, not a proven discount effect. Run an A/B test before changing discount policy.

PROFIT MODEL

Price and quantity drive profit

Linear baseline

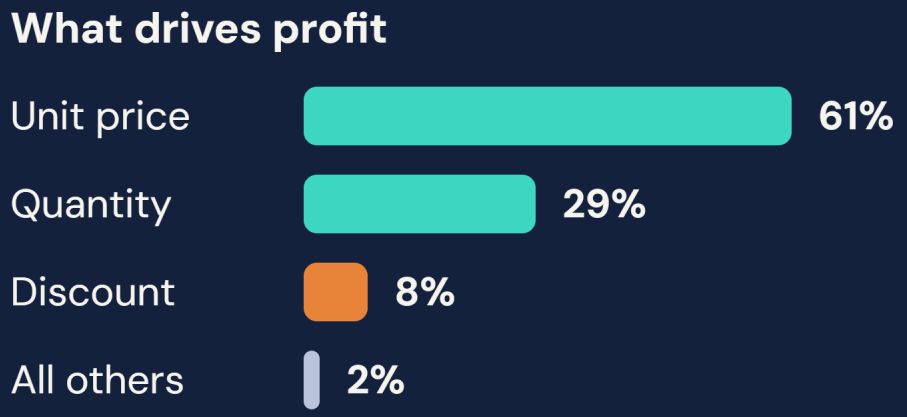
0.73

R^2

Random Forest

0.998

R^2



Read with care: profit is arithmetic on price, quantity, discount and cost, so a near-perfect fit is expected. The value is in ranking the drivers: category, region and payment method barely matter.

POWER BI

One dashboard for the whole story

KPI cards

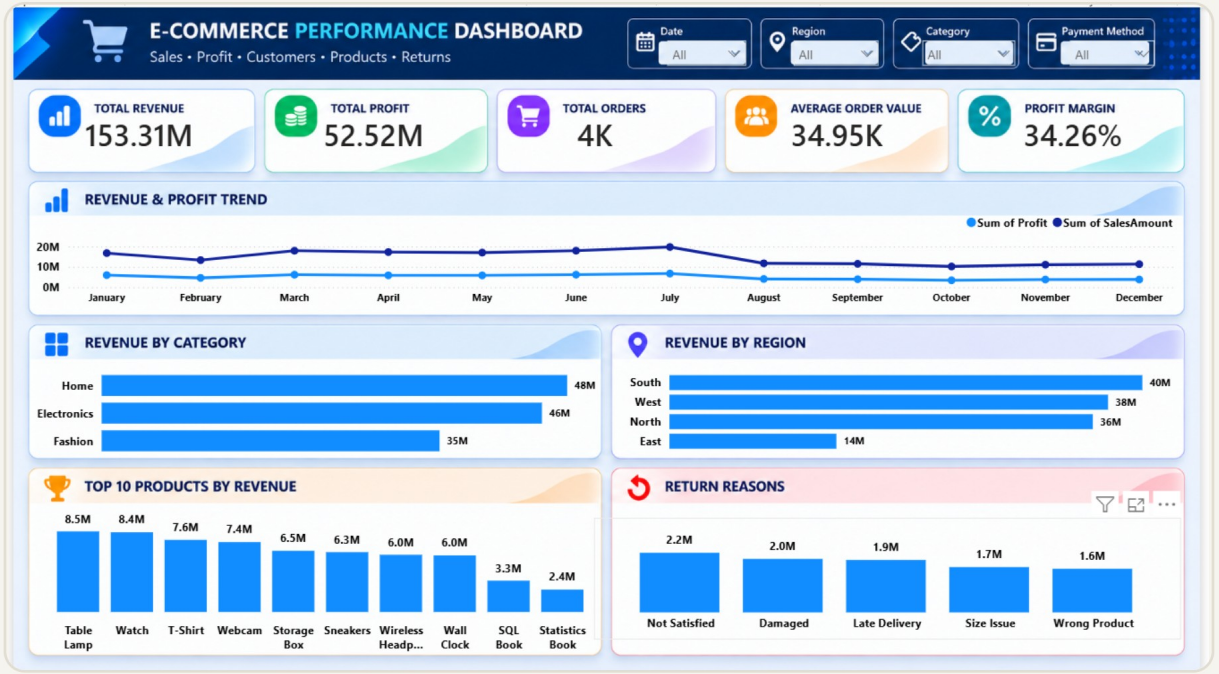
Revenue, profit, orders, profit margin %, average order value

Visuals

Monthly revenue and profit trend, revenue by category, region and product, return reasons

Slicers

Month, region, category and payment method



RECOMMENDATIONS

Five actions for the business

1. Cap discounts at 15%, or require approval above it
2. Promote high-margin lines such as Books; review Beauty pricing and costs
3. Run a targeted acquisition push in the East region
4. Cut late-delivery and wrong-item returns with carrier SLAs and warehouse QC
5. A/B test discounts before changing policy

E-COMMERCE SALES & PROFITABILITY ANALYTICS

Thank you

Questions and feedback are welcome.

Sanjeev

Code, SQL, notebooks and dashboard on GitHub:

github.com/sanjeevkushwaha78/E-Commerce-Sales-Profitability-Analytics